SRI SAI RAM ENGINEERING COLLEGE

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STUDENT ID/ REG NO:

DEPARTMENT OF CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Digital Dynamics

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ACADEMIC YEAR 2024-2025

SRI SAI RAM ENGINEERING COLLEGE

Sai Leo Nagar, West Tambaram, Chennai-600 044.

Department of CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Bonafide Certificate

REGISTER NUMBER

Certified to be the bonafide record of the work done by

_____ Student Id/ Register No. _____ of

B.E CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) in the academic year 2024-2025.

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Staff In-charge **Head of the Department**

Submitted for the SECOND Semester Practical Examination held on _____ at Sri Sai Ram Engineering College, Chennai-600 044.

Internal Examiner

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Sl. No.	DATE	TITLE	PG.No.	SIGNATURE
1		Write 25 Cyber Terminologies		
2		(Power Point Presentation select topic)		
3		Importance of Standards and Insightfulness- present an essay.		

25 Words from Cyber Terminology :

1. Quishing : Phishing attack using QR codes to lure users into malicious websites.

2. Credential Stuffing : Using stolen username/password combos to access multiple accounts.

3. Polymorphic Malware : Malware that constantly changes its code to evade detection.

4. Typosquatting : Creating fake websites with URL typos to trick users

5. Watering Hole Attack : Infecting a website that a specific group frequently visits.

6. Clickjacking : Hiding malicious links under clickable content to trick users.

7. Fileless Malware : Malware that operates in memory, avoiding traditional file-based detection.

8. Skimming : Stealing card data through physical or virtual means

9. Key Escrow : A system where encryption keys are held by a third party for lawful access.

10. Rootkit : A stealthy set of malicious tools used to gain root access.

11. Logic Bomb : Code that triggers a malicious function when specific conditions are met.

12. Zero Trust Architecture : Security model assuming no implicit trust inside or outside the network.

13. Cyber Hygiene : Best practices for maintaining digital security health.

14. Cyber Deception : Using decoys and traps to detect and mislead attackers.

15. Supply Chain Attack : Infiltrating systems via vulnerabilities in third-party vendors.

16. Man-in-the-Browser : Malware that intercepts data between the browser and user.

17. Juice Jacking : Data theft via public USB charging stations.

18. Rogue Access Point : Unauthorized Wi-Fi device used to eavesdrop or attack networks.

19. Doxing : Publishing personal information about someone with malicious intent.

20. Air Gap Attack : Breaching isolated systems using indirect methods

21. Bug Bounty : Reward offered for discovering and reporting security vulnerabilities.

22. Data Exfiltration : Unauthorized transfer of data from a device or network.

23. Threat Hunting : Proactively searching for hidden cyber threats in a system.

24. Session Hijacking : Taking control of a user's active web session.

25. Smishing : Phishing via SMS to steal credentials or deliver malware.

The Importance of Digital Ethics

Embracing Responsibility in an Evolving Digital Landscape

What is Digital Ethics?

Digital ethics explores how technology shapes our lives. It emphasizes responsible digital tool use. It ensures innovation aligns with moral values.

1

Technology's Impact

Examines how tech influences behavior.

2

Responsible Use

Focuses on ethical digital practices.

3

Moral Alignment

Ensures innovation respects values.

Core Values of Digital Ethics

Four pillars form the foundation of digital ethics. These values guide our interaction with technology.

Privacy

Respecting personal data and user autonomy.

Transparency

Open, honest communication about data use.

Accountability

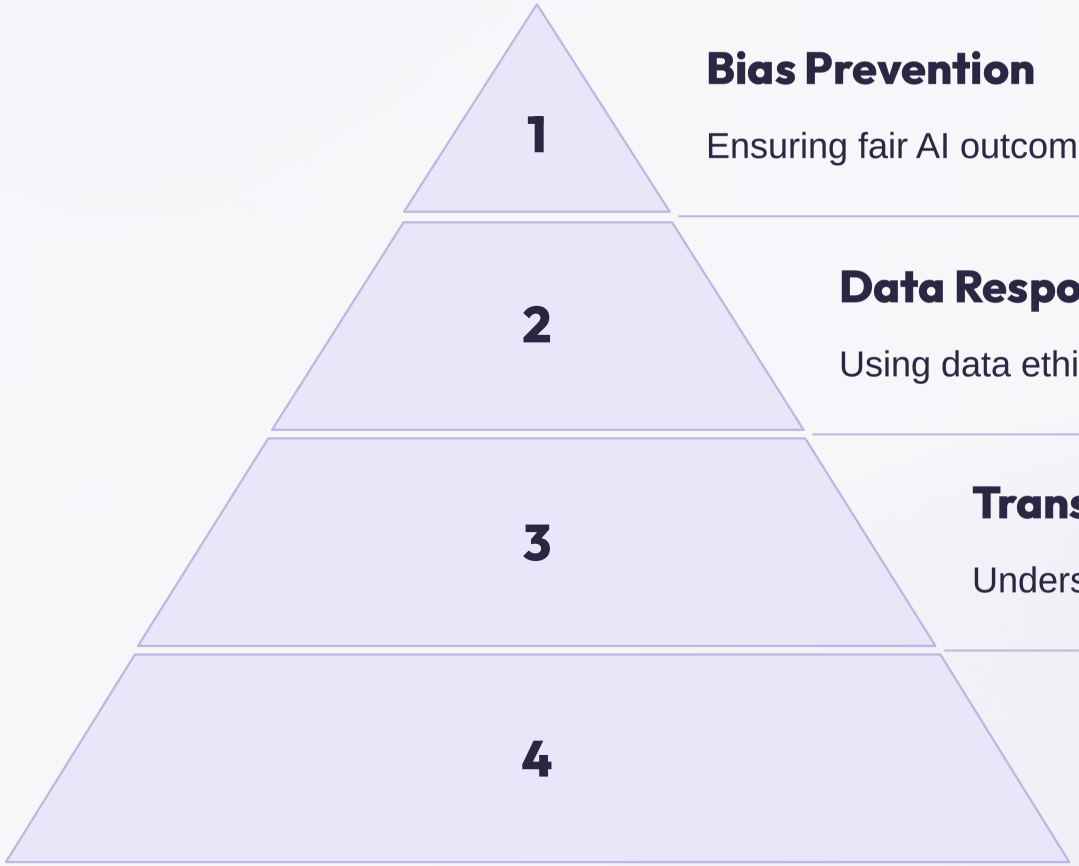
Owning consequences of tech decisions.

Fairness

Designing for inclusion and impartiality.

Why It Matters in AI & Machine Learning

AI can significantly impact real lives. Ethics ensure responsible data use and fair decisions. It prevents bias and harm.



Bias Prevention

Ensuring fair AI outcomes.

Data Responsibility

Using data ethically in AI.

Transparency

Understanding AI decision-making.

Societal Impact

Minimizing harm, maximizing good.

Challenges in Social Media & Cybersecurity

Digital spaces face misinformation and cyberbullying. Data leaks are also a major concern. Ethical design is crucial for a trustworthy internet.

Misinformation

Combating false information spread.

Cyberbullying

Protecting users from online harassment.

Data Leaks

Securing sensitive personal information.

Role of Corporates & Government

Governments regulate, while companies execute. Ethical policies are vital to protect society. This includes GDPR and responsible AI.

Government

- Establishes regulations (e.g., GDPR)
- Sets legal frameworks for tech
- Ensures public protection

Corporates

- Implements ethical policies
- Practices transparent data handling
- Develops responsible AI systems

The Future of Ethical Tech

Ethics must be embedded in future innovation. New technologies like metaverse and biotech need human values. This ensures a responsible future.

Integrate Ethics

Embed ethical considerations into all tech development phases from the start.

Guide New Tech

Apply human values to emerging areas like AI, metaverse, and biotech for responsible growth.

Foster Dialogue

Encourage ongoing conversations about tech's societal impact and ethical implications.

Conclusion – A Shared Responsibility

Digital ethics is everyone's duty. Engineers, leaders, and users must collaborate. This collaboration builds a fair, safe, and just tech future.

Users

Advocate for privacy and responsible use.

Engineers

Design ethical, unbiased systems.

Leaders

Implement and enforce ethical policies.

Society

Collaborate for a just digital future.

Bureau of Indian Standards

The Importance of Standards (BIS) and Insightfulness

In today's fast-paced world, where technology advances rapidly and industries expand across borders, standards play a crucial role in ensuring quality, safety, and trust. They help create a reliable framework for manufacturing, infrastructure, and consumer protection. In India, the Bureau of Indian Standards (BIS) serves as the national authority responsible for setting and maintaining these standards. However, beyond its technical role, BIS must embrace insightfulness—the ability to foresee future challenges and opportunities—to remain relevant in an evolving global landscape.

Standards: The Foundation of Trust

Standards are the invisible yet essential backbone of modern civilization. They ensure that bridges are structurally sound, drinking water is safe, and electronic devices function properly. By providing uniformity, reliability, and safety, BIS helps industries build products that meet expectations, while also protecting consumers from unsafe goods. Standards do not merely regulate industries; they create trust, assuring businesses and individuals that the products they use daily are safe and effective.

BIS: More Than Just a Certifying Body

The Bureau of Indian Standards is not just a government agency—it is a benchmark of excellence that shapes industries and national growth. BIS plays a key role in:

- Boosting Indian Manufacturing: By ensuring that ‘Made in India’ products meet both national and international benchmarks, BIS helps Indian industries compete globally.

- Protecting Consumer Interests: By enforcing safety measures, BIS prevents hazardous products from reaching the market and promotes ethical manufacturing.

- Strengthening Infrastructure: BIS provides standards for crucial sectors such as smart cities, clean energy, and digital transformation, ensuring they develop efficiently and sustainably.

- Facilitating International Trade: By aligning Indian standards with global requirements, BIS helps reduce technical barriers in exports, making Indian goods more competitive in global markets.

Through these initiatives, BIS creates a strong foundation for innovation, allowing industries to grow while maintaining high standards of quality and safety.

The Power of Insightfulness in Standardization

While standards provide a framework for the present, insightfulness ensures preparedness for the future. A forward-thinking approach allows BIS to anticipate emerging technological, social, and economic challenges. This involves:

- Adapting to Technological Advancements: With innovations in AI, quantum computing, and smart technologies, BIS must craft standards that keep India at the forefront of technological progress.

- Balancing Innovation with Ethics: Growth should not come at the cost of environmental or societal well-being. Insightfulness helps BIS develop standards that protect the planet and communities while encouraging business expansion.

- Connecting Different Industries: Standards in one area, such as data security, can significantly impact other sectors like healthcare and fintech. BIS must recognize these interconnections and create holistic guidelines.

- Supporting National Policies: Initiatives such as Atmanirbhar Bharat, Smart Cities, and the Sustainable Development Goals (SDGs) benefit from well-crafted standards that align with India's long-term development goals.

Through insightfulness, BIS ensures that its standards are not rigid rules of the past but flexible guidelines for the future, promoting innovation while maintaining safety.

Standards: A Blueprint for Progress

In the data-driven era, standards have evolved beyond technical regulations—they shape how nations design industries, build infrastructure, and compete globally. BIS must remain agile and responsive, ensuring that India is not just following global trends but leading them.

A strategic approach to standardization helps India strengthen its industries, protect its citizens, and enhance its global presence.

Conclusion: Building India’s Future with Standards

A country without strong standards faces uncertainty and risk. A country with rigid standards but no vision stagnates. However, a country that combines clear regulations with forward-thinking insight builds not just industries—it builds legacies.

The Bureau of Indian Standards plays a critical role in shaping India's future. By upholding quality, innovation, and foresight, BIS ensures that India's industries thrive, its infrastructure advances, and its consumers remain protected. More than a regulatory body, BIS stands as a pillar of progress, defining the standards of India's growth and global leadership.